TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University:

Pimpri Chinchwad College of Engineering (PCCOE)

Programme / Batch:

CS / AIML

Student Name:

Ayush Tatya Ahire

PRN:

122B1C001

1. Case Study ID and Project Title / Topic

Case Study ID (CS1 to CS5):

CS3

Project Title / Topic:

Automotive Cybersecurity TARA Assistant

2. Objectives

Define the specific objectives and focus areas of the allocated AIML case study.

Objectives

- Implement an AI-assisted Threat Analysis and Risk Assessment (TARA) tool conforming to ISO/SAE 21434.
- Automate asset extraction and threat scenario generation using Large Language Models.
- Ground LLM outputs using a deterministic RAG pipeline backed by a UN R155 regulatory threat library.
- Evaluate risks using a standardized risk matrix and trace mitigations directly to threats.

3. Scope

Clearly state what will be implemented in the project and what will not be covered.

In Scope - To Be Implemented

- Architecture JSON parsing and trust boundary mapping.
- RAG-based threat and mitigation retrieval using ChromaDB.
- FastAPI backend serving LLM integration with fallback chains (Mistral -> Groq).
- Human-in-the-loop review UI via Streamlit.

Out of Scope - Will Not Be Implemented

- Direct execution of penetration tests or automated code scanning.
- Live vehicle telemetry ingestion or active monitoring.
- Non-regulatory threat libraries not specified in UN R155.

4. Proposed Approach

Pre-trained model (name, version, local/private hosting)

Mistral (mistral-small-latest) / Groq LLaMA-3 (fallback via API)

Embedding model

BAAI/bge-small-en-v1.5

Vector store (FAISS / ChromaDB)

ChromaDB

RAG pipeline and/or AI-agent design - brief description

LangGraph state machine orchestrating 'Knowledge' (deterministic retrieval) and 'Analyst' (LLM drafting) nodes.

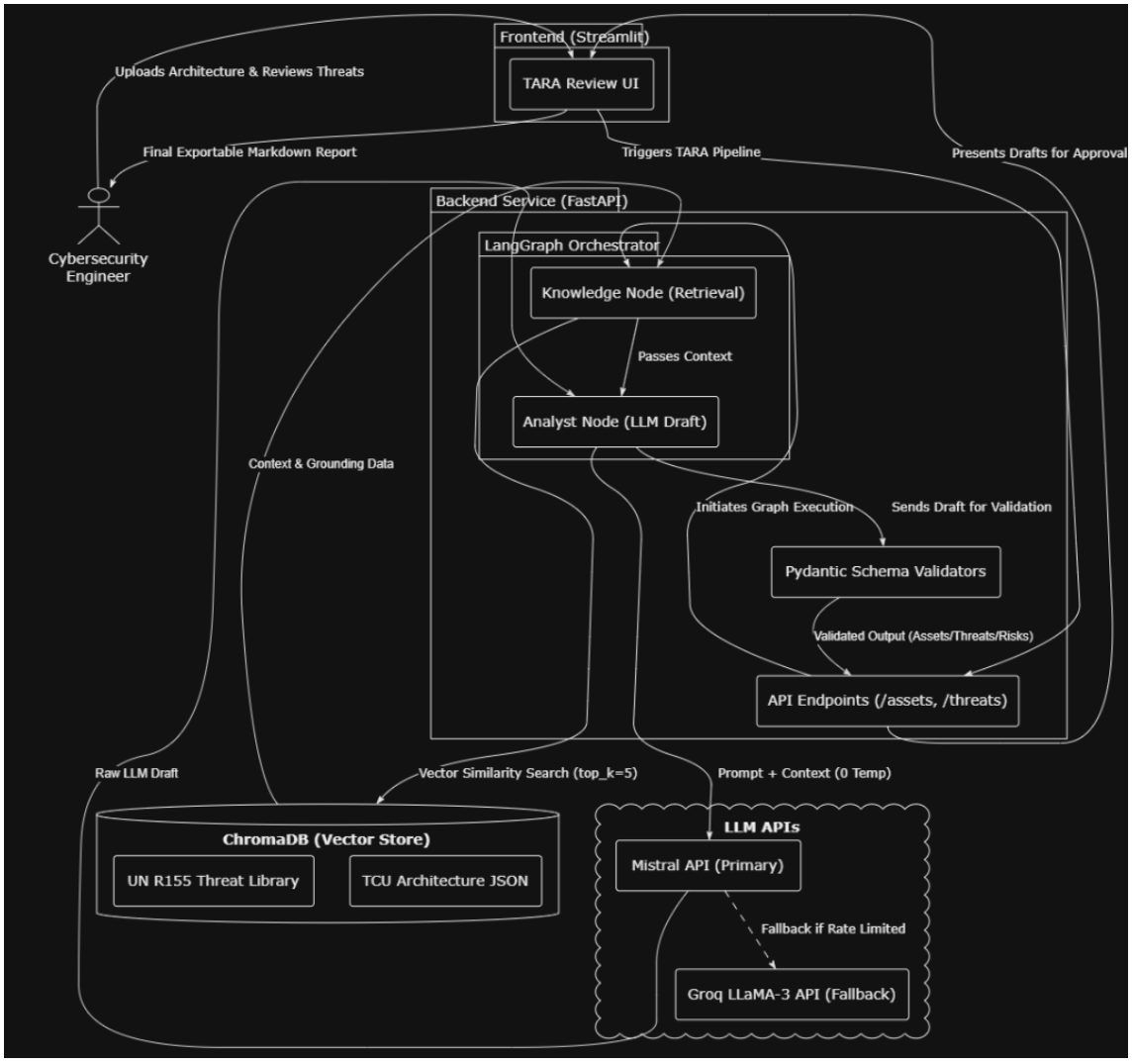
Service layer and UI (e.g., FastAPI / Flask / Streamlit)

FastAPI (Backend) / Streamlit (UI)

Tools / deterministic rules, if applicable

Strict JSON schema enforcement via Pydantic; deterministic risk computation via 5x5 matrix.

Proposed System / Architecture



5. Input Data / Knowledge Base

Use approved or synthetic data only.

Data / Knowledge Sources

JSON architecture files, UN R155 threat library, asset catalogue.

Approximate Size and Format

JSON arrays (~3000 lines, <1MB).

Preprocessing and EDA Planned

Flattening interfaces, mapping STRIDE categories, mapping trust boundaries.

6. Expected Outcomes and Evaluation Plan

Expected Outputs - aligned to the case study

A fully traceable TARA report in Markdown with assets, threats, risks, and mitigations.

Evaluation Method - test question set / ground truth / metrics / validation approach

Human review pass/fail metrics; Traceability gaps metric.

Responsible AI Measures - grounding and citations / privacy / security / human oversight

All AI drafts require explicit human approval via the UI. Strict RAG citations. Only synthetic data used.

MLOps / Deployment Elements Planned - versioning / Docker / tracking

FastAPI backend, standalone execution via run.bat.

7. Planned AI-Tool Usage

Mention the AI tools / models used for coding assistance and specify the purpose and project artifact affected.

AI Tool / Model

Gemini / Antigravity AI

Purpose

Coding assistance, architecture migration, debugging.

Artifact Affected

Codebase and documentation.

Student Declaration

[X] I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

[X] I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Name and Signature: Ayush Tatya Ahire

Date: 07 Oct 2026